

Database Principles: Fundamentals of Design, Implementation, and Management

Lecture 1: Database Introduction

Updated: September 2025

Learning Outcomes

After completing this lecture, students will be able to:

- Differentiate between **data** and **information**
- Define **database**, describe **types**, and explain their value **for decision-making**
- Explain the **evolution of databases** from manual and computerized file systems
- Identify **flaws in file system** data management
- Describe main **components of a database system**
- Explain key **functions of a DBMS**
- Explain **Systems Development Life Cycle (SDLC)** in developing successful information systems.
- Recognize the importance of **Database Life Cycle (DBLC)** for evaluating and revising databases

Why Databases?

- Databases **solve many of the problems** encountered in data management
 - Used in **almost all modern settings** involving data management:
 - Business
 - Research
 - Administration
- Important to understand **how databases work** and **interact with other applications**

Data vs. Information

Data	Information
1. Raw fact (e.g. name, person, thing etc) 2. Must be formatted for processed, storage and presentation 3. Building blocks of information	1. Results of processed data (e.g. student details, course details etc) 2. Used to reveal meaning of data 3. Used for decision making

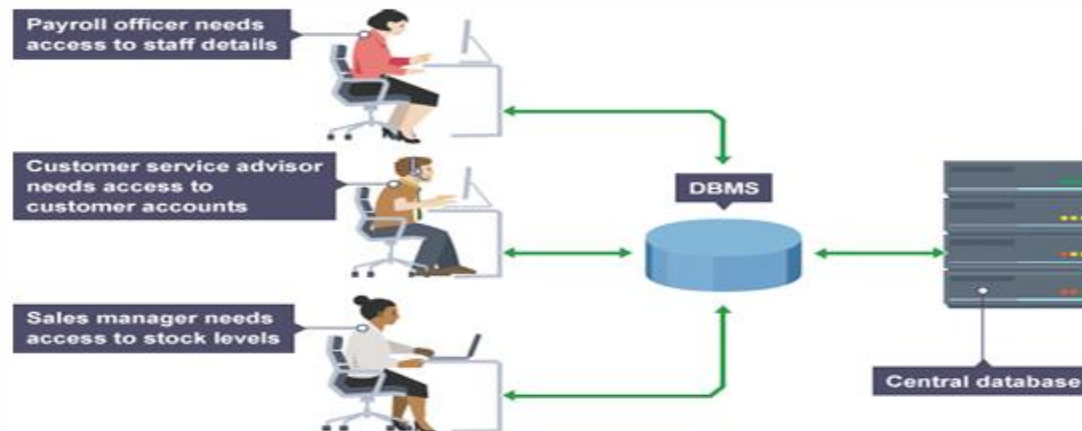
Introducing the Database

- **Database:** **shared, integrated computer structure** that **stores** a collection of:
 - **End-user data:** **raw facts** of interest to end user
 - **Metadata:** **data about data**
 - Provides description of **data characteristics** and **relationships in data**
- **Database management system (DBMS):** **collection of programs**
 - **Manages** structure and controls access to data



Role and Advantages of the DBMS

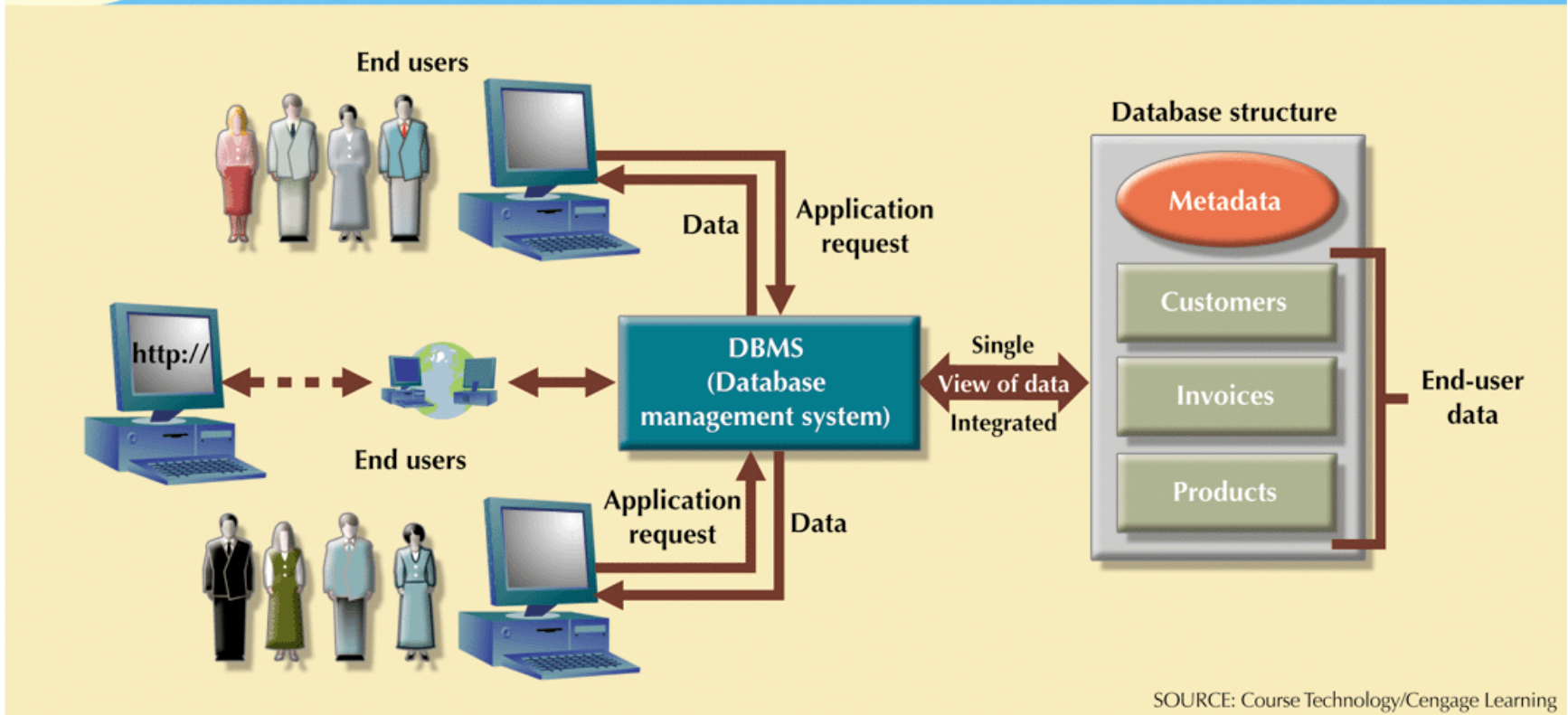
- DBMS is the **intermediary** between the user and the database
 - **Database structure** stored as file collection
 - Can only **access files** through the DBMS
- DBMS enables **data to be shared**
- DBMS **integrates many users' views** of the data



Role and Advantages of the DBMS (cont'd.)

FIGURE 1.2

The DBMS manages the interaction between the end user and the database



Role and Advantages of the DBMS (cont'd.)

- **Advantages of a DBMS:**
 - Improved **data sharing**
 - Improved **data security** – eg: password policies
 - Better **data integration** – eg: HR department data + Faculties data
 - Minimized **data inconsistency** – eg: less data appear in different places
 - Improved **data access** – eg: quick answers to ad hoc queries
 - Improved **decision making**
 - Increased end-user **productivity**

Types of Databases

- Databases can be **classified** according to:
 - Number of **users**
 - Database **location(s)**
 - Expected type and extent of **use**

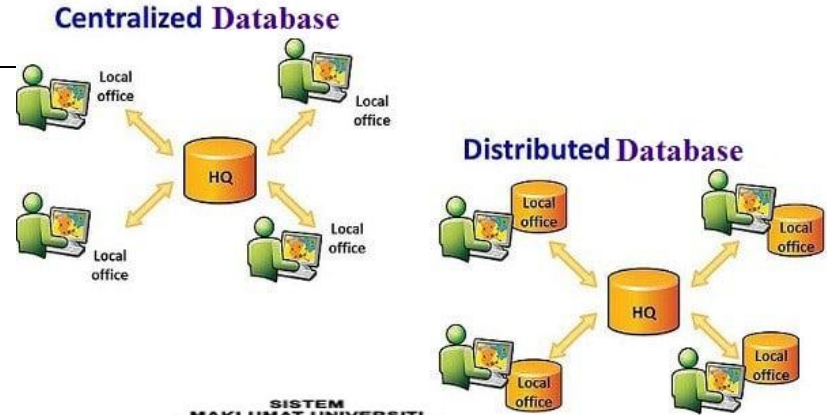
1. Number of users:

- **Single-user database** supports only **one user at a time**
 - **Desktop database**: single-user; runs on PC
- **Multiuser database** supports **multiple users at the same time**
 - **Workgroup and enterprise databases**
 - **Workgroup database** (supports less than 50 users at the same time)
 - **Enterprise database** (supports more than 50 users at the same time)

Types of Databases (cont'd.)

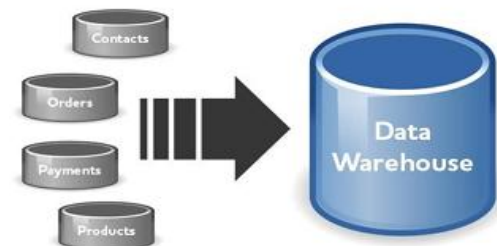
2. Database location(s):

- **Centralized database:** data located at a **single site**
- **Distributed database:** data distributed across **several different sites**



3. Expected type and extent of use:

- **Operational database:** supports a company's **day-to-day operations**
 - Transactional or production database
- **Data warehouse:** stores data used for **tactical or strategic decisions**

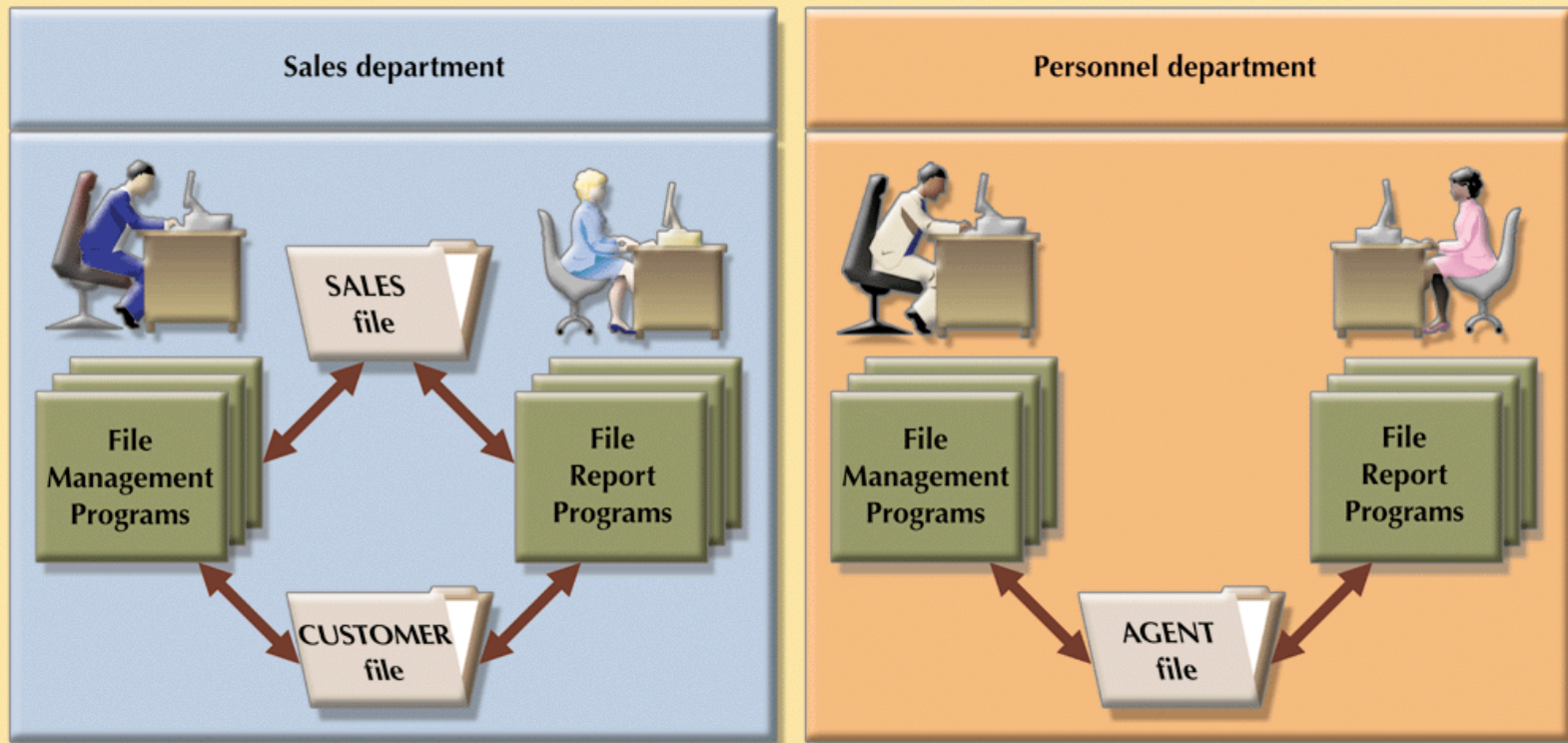


File System Data Processing - Modern End-User Productivity Tools

- Use of **personal productivity tools** can introduce the same problems as the old file systems
- **Microsoft Excel**
 - **Widely used** by business users
 - Users have become so adept at working with spreadsheets, they tend to use them **to complete tasks** for which spreadsheets are not appropriate – database substitute

**FIGURE
1.7**

A simple file system



SOURCE: Course Technology/Cengage Learning

Problems with File System Data Processing

- **Summary of file system limitations:**
 - Cannot perform **ad hoc queries**
 - System administration is **complex and difficult**
 - **Difficult to make changes** to existing structures
 - **Security features** are likely to be inadequate

Data Redundancy

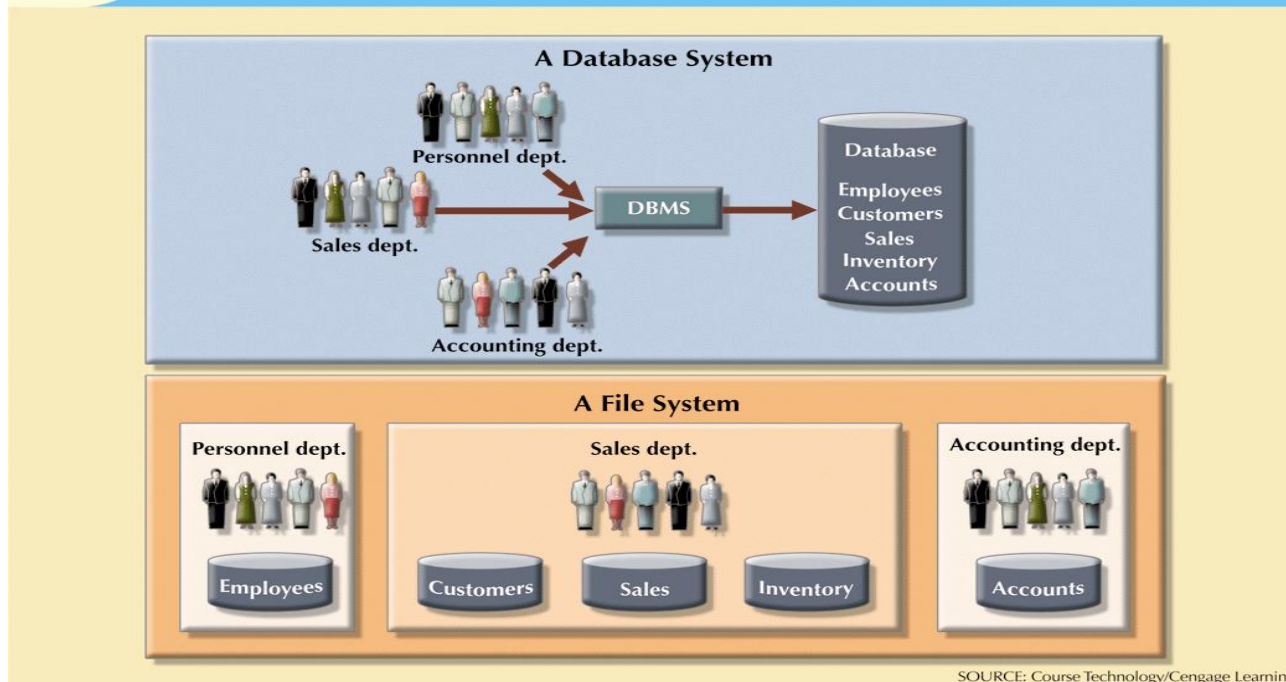
- File system structure makes it **difficult to combine data** from multiple sources
- Organizational structure promotes **storage of same data in different locations**
- **Data redundancy:** **same data stored unnecessarily in different places**
- **Data inconsistency:** different and **conflicting versions of same data** occur at different places
- **Data anomalies:** abnormalities when all **changes in redundant data are not made correctly**
 - **Update anomalies**
 - **Insertion anomalies**
 - **Deletion anomalies**

Database Systems

- Database system consists of **logically related data stored in a single logical data repository**

FIGURE 1.8

Contrasting database and file systems



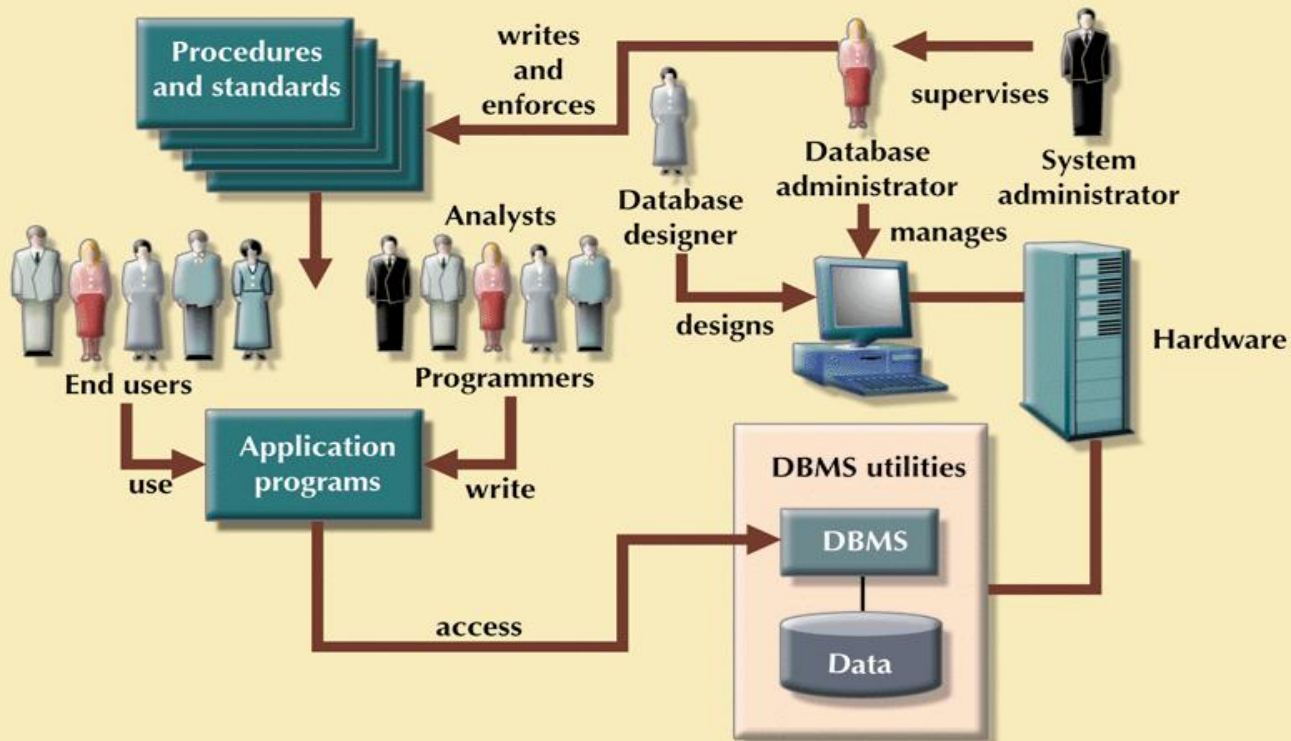
The Database System Environment

- **Database system:** defines and regulates the collection, storage, management, use of data
- **Five major parts of a database system:**
 - **Hardware**
 - **Software**
 - **People**
 - **Procedures**
 - **Data**

The Database System Environment (cont'd.)

FIGURE 1.9

The database system environment



SOURCE: Course Technology/Cengage Learning

The Database System Environment (cont'd.)

- **Hardware:** all the system's **physical devices**
- **Software:** three types of software required
 - **Operating system** software
 - **DBMS** software
 - **Application programs** and **utility software**
- **People:** **all users** of the database system
 - System and database **administrators**
 - Database **designers**
 - Systems analysts and **programmers**
 - End **users**
- **Procedures:** **instructions and rules** that govern the design and use of the database system
- **Data:** the **collection of facts** stored in the database

DBMS Functions

- Most functions are transparent to end users
 - Can only be achieved through the DBMS
- **Data dictionary management**
 - DBMS stores definitions of data elements and relationships (metadata) in a data dictionary
 - DBMS looks up required data component structures and relationships
 - Changes automatically recorded in the dictionary

**FIGURE
1.10**

Illustrating metadata with Microsoft SQL Server Express

The screenshot displays the Microsoft SQL Server Management Studio Express interface. The Object Explorer on the left shows the database structure, including the 'dbo.CUSTOMER' table. The Table Designer on the right shows the table's columns and properties. A red box highlights the table's columns, and another red box highlights the column properties for 'C_NAME'. A red arrow points from the word 'Metadata' to these areas.

Column Name	Data Type	Allow Nulls
C_NAME	varchar(20)	☒
C_PHONE	varchar(12)	☒
C_ADDRESS	varchar(30)	☒
C_ZIP	varchar(5)	☒
A_NAME	varchar(20)	☒
A_PHONE	varchar(12)	☒
TP	varchar(2)	☒
AMT	numeric(6, 2)	☒
REN	datetime	☒

Column Properties

(General)

(Name)	C_NAME
Allow Nulls	Yes
Data Type	varchar
Default Value or Binding	
Length	20

Table Designer

Collation	<database default>
Computed Column Specification	
Condensed Data Type	varchar(20)
Description	
Deterministic	Yes
DTS-published	No
Full-text Specification	No
Has Non-SQL Server Subscriber	No
Identity Specification	No

Metadata

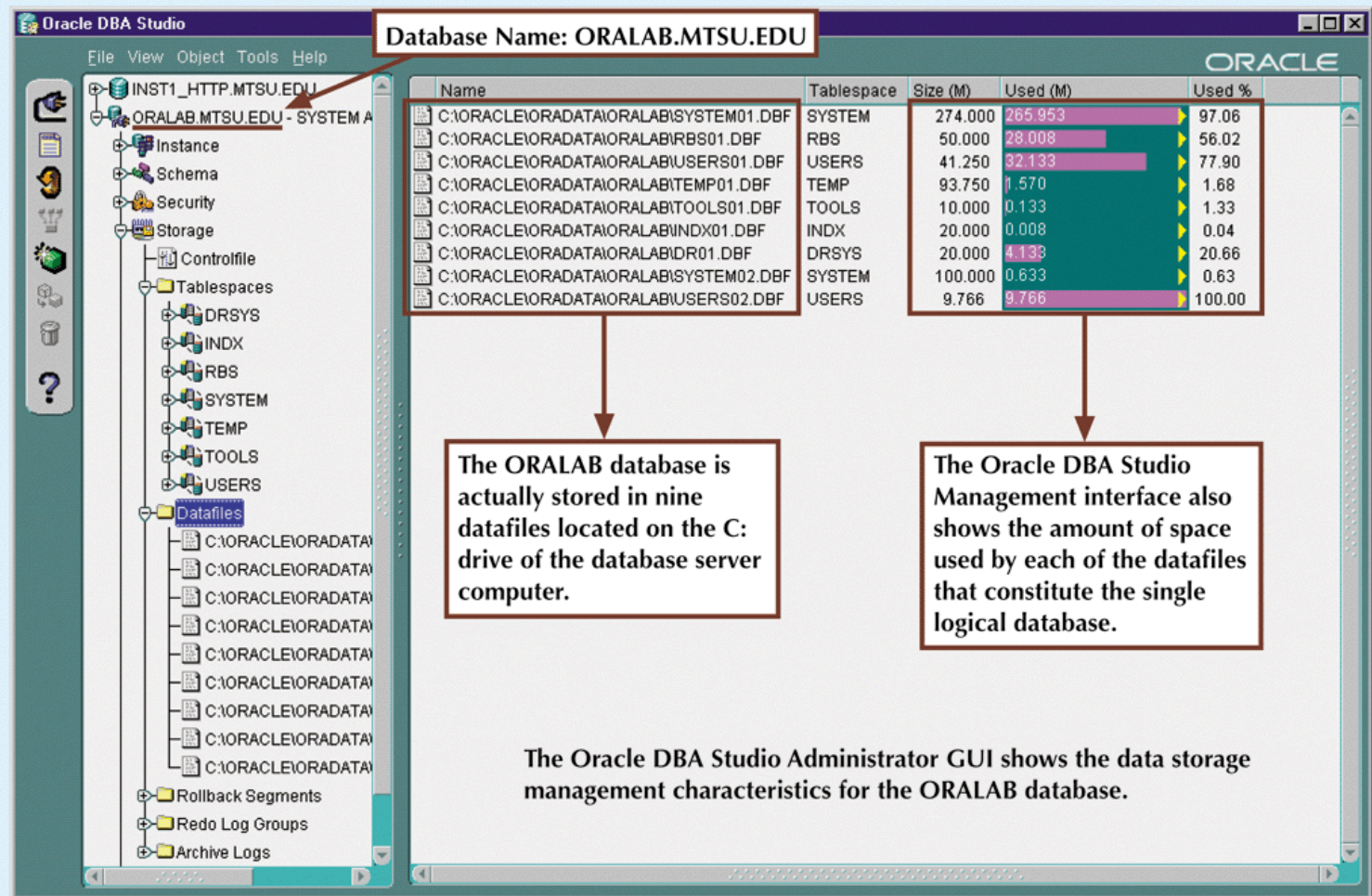
SOURCE: Course Technology/Cengage Learning

DBMS Functions (cont'd.)

- **Data storage management**
 - DBMS **creates and manages complex structures** required for data storage
 - Also **stores related** data entry forms, screen definitions, report definitions, etc.
 - DBMS stores the database in **multiple physical data files**

FIGURE 1.11

Illustrating data storage management with Oracle



SOURCE: Course Technology/Cengage Learning

DBMS Functions (cont'd.)

- **Data transformation and presentation**
 - DBMS **transforms data entered** to conform to required data structures
 - DBMS **transforms physically retrieved data** to conform to user's logical expectations
- **Security management**
 - DBMS creates a security system that **enforces user security and data privacy**
 - **Security rules** determine which users can access the database, which items can be accessed, etc.
- **Multiuser access control**
 - DBMS uses sophisticated algorithms to ensure **concurrent access** does not affect integrity

DBMS Functions (cont'd.)

- **Backup and recovery management**
 - DBMS provides **backup and data recovery** to ensure data safety and integrity
 - Recovery management deals with recovery of database **after a failure**
- **Data integrity management**
 - DBMS promotes and **enforces integrity rules**
 - **Minimizes** redundancy
 - **Maximizes** consistency
 - **Data relationships** stored in data dictionary used to enforce data integrity

DBMS Functions (cont'd.)

- **Database access languages and application programming interfaces**
 - DBMS provides access through a **query language**
 - Query language is a **nonprocedural language**
 - **Structured Query Language (SQL)** is the de facto query language
- **Database communication interfaces**
 - Current DBMSs accept end-user requests via **multiple different network environments**

Disadvantages of Database Systems

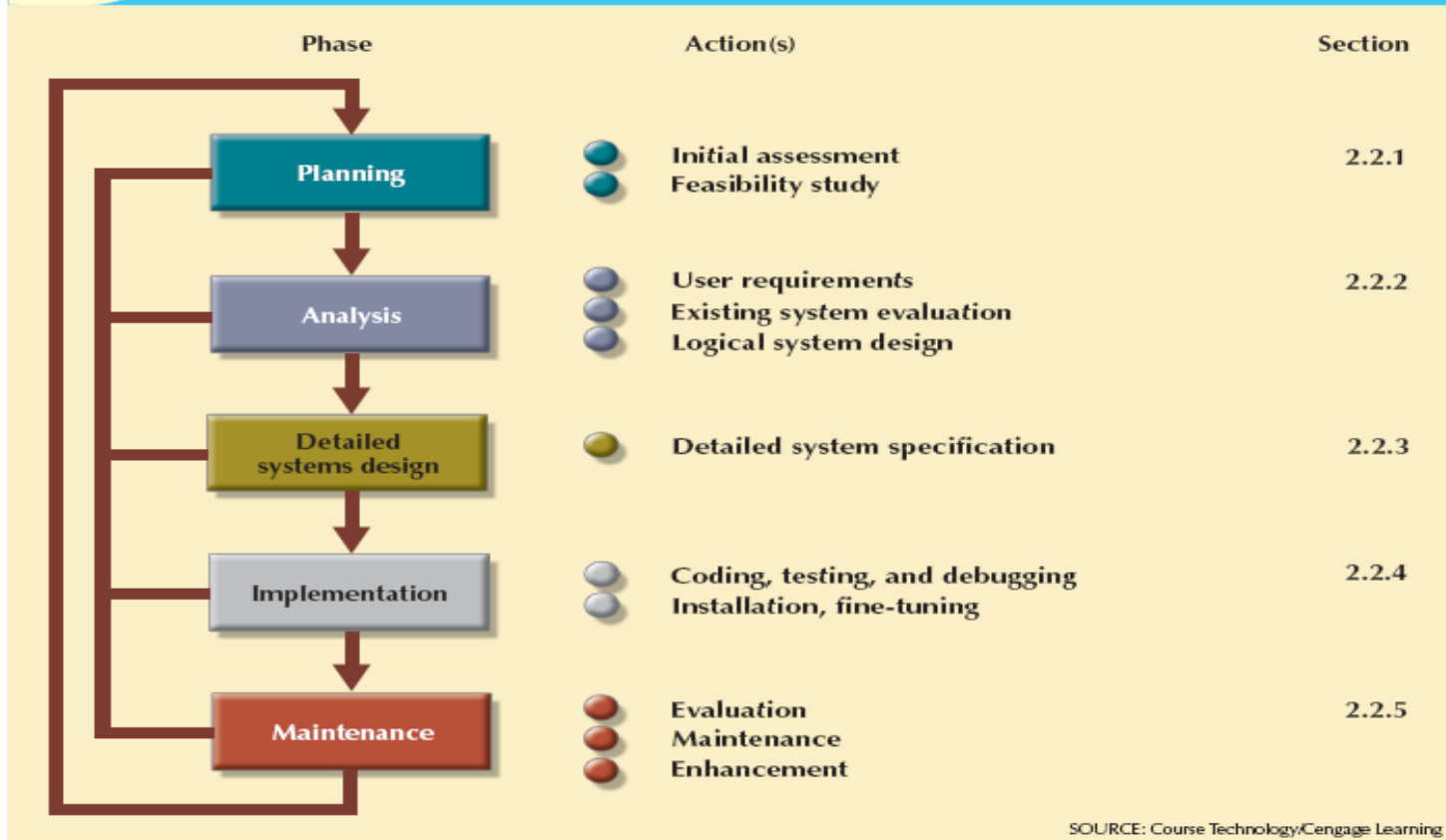
Database systems offer many advantages; however, they also have **some disadvantages**, as follows:

- **Increased costs** – hardware, software, highly skilled personnel, training etc.
- **Management complexity** – many different technologies.
- **Maintaining currency** – must keep your system current.
- **Vendor dependence** – heavy investment in technology and personnel training, companies might be reluctant to change database vendors.
- **Frequent upgrade/replacement cycles** – DBMS vendors frequently upgrade their products by adding new functionality. Some of the versions require hardware upgrades. Cost money to train database users and administrators to properly use and manage the new features.

The Systems Development Life Cycle (SDLC)

FIGURE 2.2

The Systems Development Life Cycle



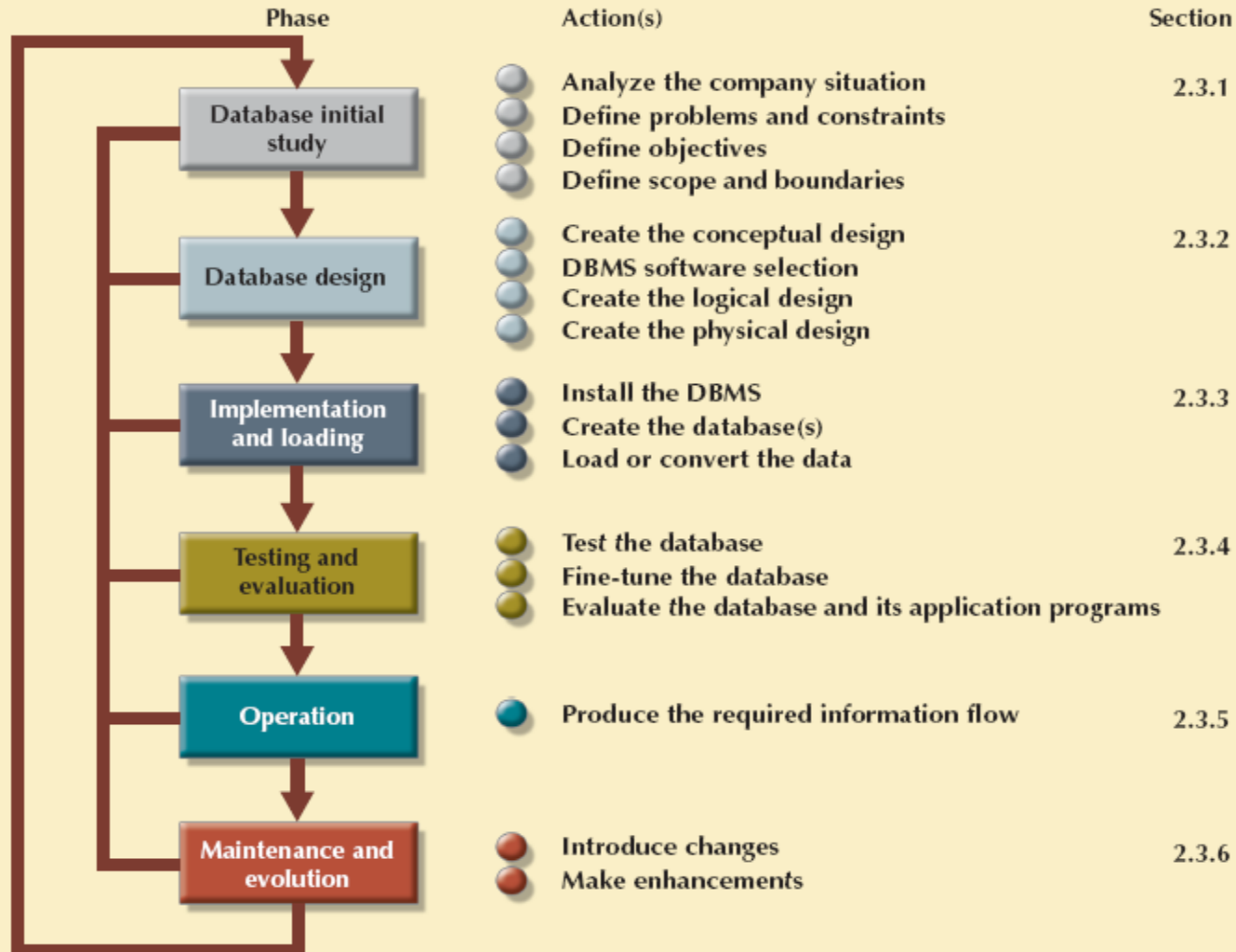
SOURCE: Course Technology/Cengage Learning

The Database Life Cycle (DBLC)

- **Six phases:**
 - Database initial study
 - Database design
 - Implementation and loading
 - Testing and evaluation
 - Operation
 - Maintenance and evolution

**FIGURE
2.3**

The Database Life Cycle



SOURCE: Course Technology/Cengage Learning

The Database Initial Study

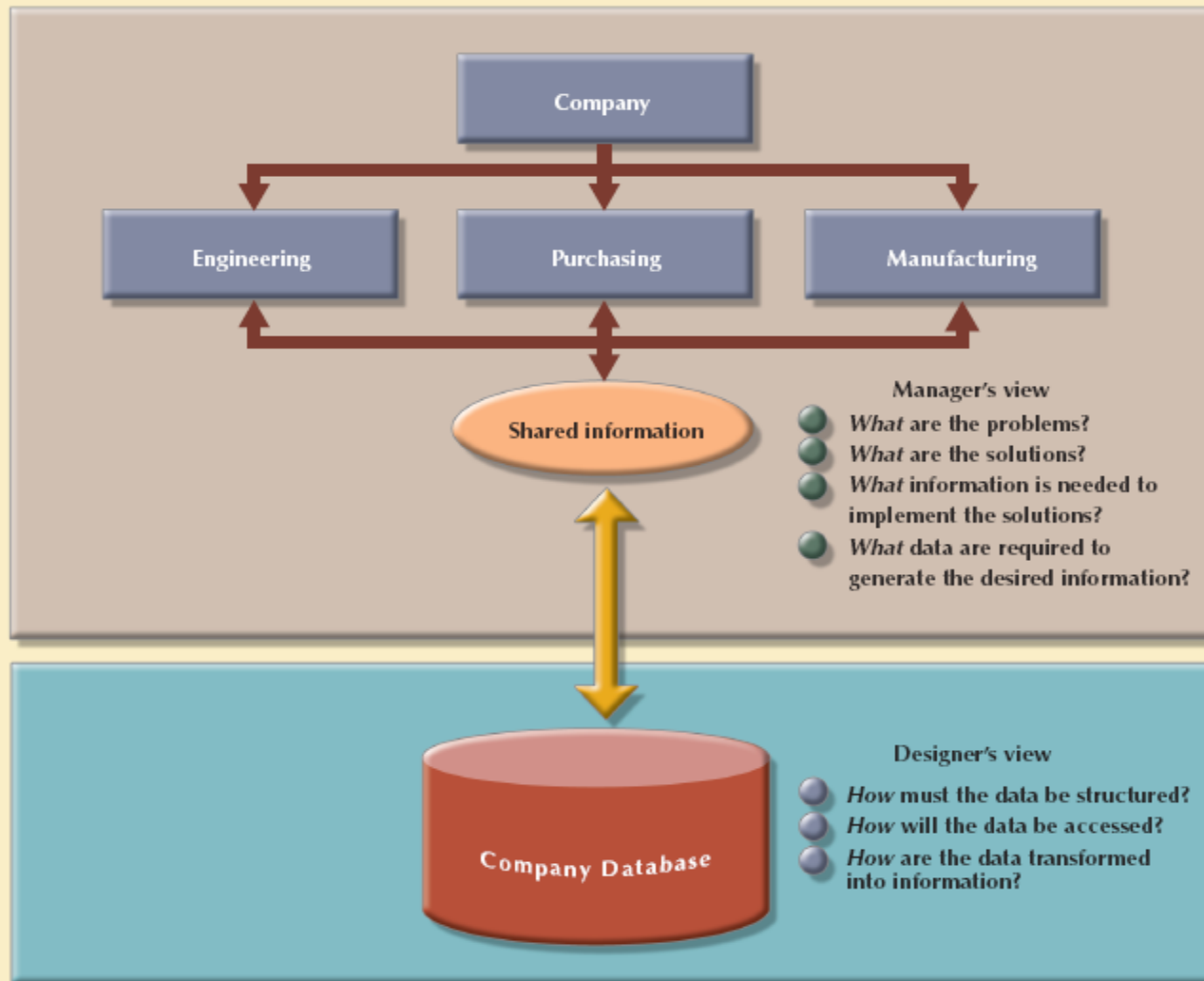
- **Overall purpose:**
 - Analyze **company situation**
 - Define **problems** and constraints
 - Define **objectives**
 - Define **scope** and boundaries
- **Interactive and iterative processes** required to complete first phase of DBLC successfully

Database Design

- Necessary to concentrate on **data characteristics** required to build **database model**
- **Two views of data** within system:
 - **Business view**
 - Data as information source
 - **Designer's view**
 - Data structure, access, and activities required to transform data into information

FIGURE
2.5

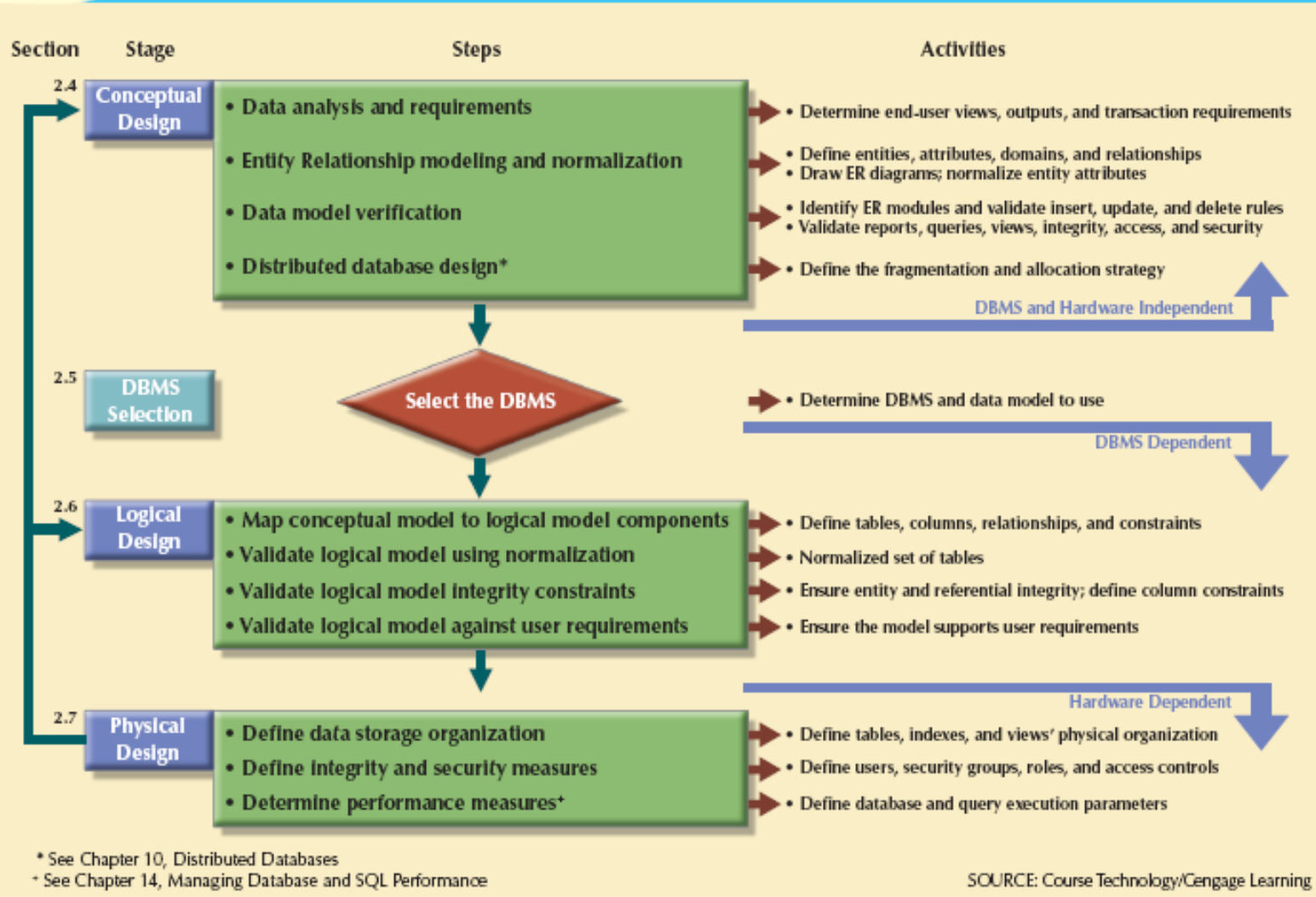
Two views of data: business manager and database designer



SOURCE: Course Technology/Cengage Learning

FIGURE 2.6

Database design process

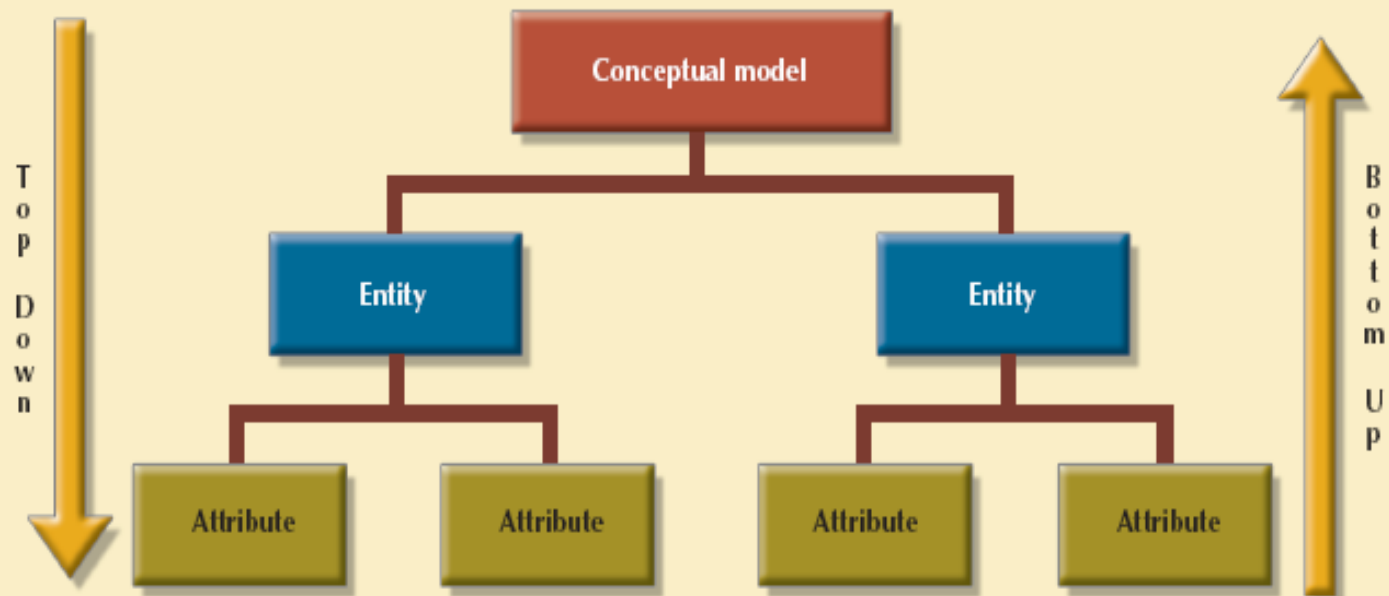


Database Design Strategies

- **Top-down design (Lecture 3 & 4)**
 - Identifies data sets
 - Defines data elements for each of those sets
 - Definition of different entity types
 - Definition of each entity's attributes
- **Bottom-up design (Lecture 5 & 6)**
 - Identifies data elements (items)
 - Groups them together in data sets

**FIGURE
2.14**

Top-down vs. bottom-up design sequencing



SOURCE: Course Technology/Cengage Learning

Implementation and Loading

- **Implement all design specifications** from previous phase:
 - **Install the DBMS**
 - **Create the Database**
 - **Load or Convert the Data**

Testing and Evaluation

- Occurs in **parallel with applications programming**
- **If implementation fails** to meet some of system's evaluation criteria:
 - **Fine-tune** specific system and DBMS configuration parameters
 - **Modify** physical or logical design
 - **Upgrade** software and/or hardware platform
- **Integrity:** Enforced via proper use of primary, foreign key rules
- **Backup and Recovery:**
 - Full backup
 - Differential backup
 - Transaction log backup

Operation

- Once database has **passed evaluation stage**, it is considered operational
- Beginning of operational phase starts **process of system evolution**
- **Problems not foreseen** during testing surface
- **Solutions** may include:
 - **Load-balancing** software to distribute transactions among multiple computers
 - Increasing **available cache**

Maintenance and Evolution

- **Required periodic maintenance:**
 - Preventive maintenance
 - Corrective maintenance
 - Adaptive maintenance
 - Perfective maintenance
 - Assignment of access permissions and their maintenance for new and old users
 - Generation of database access statistics
 - Periodic security audits
 - Periodic system-usage summaries

Maintenance and Evolution (cont'd.)

Preventive maintenance

- Preventive maintenance involves **proactive measures** taken to detect, reduce, or prevent potential system problems before they occur.
- The goal is to **maintain system** reliability, avoid downtime, and extend the system's useful life.

Examples:

- Regularly **backing up** databases to prevent data loss.
- **Monitoring system logs** to identify early signs of failure.
- **Updating antivirus and firewall** software to prevent security breaches.
- Conducting **routine hardware checks** to ensure components are functioning properly.
- Reviewing and **revising documentation** to ensure procedures are current and accurate.

Maintenance and Evolution (cont'd.)

Corrective maintenance

- Corrective maintenance refers to the process of **identifying and fixing errors or faults** in an information system **after it has been implemented**.
- It focuses on **correcting issues** that prevent the system from functioning as intended.

Examples:

- Fixing a **bug** in a database query that returns incorrect results.
- Repairing a **login error** that prevents users from accessing the system.
- Correcting **data entry validation issues** that allow invalid data to be saved.
- Resolving **system crashes or slow performance** caused by faulty code.
- Repairing **file corruption** in the system's data storage.
- Fixing **interface display issues** (e.g., buttons not working).

Maintenance and Evolution (cont'd.)

Adaptive maintenance

- Adaptive maintenance occurs when **changes are made** to an information system **due to external factors** or strategic decisions within the organization.
- The system is modified to ensure it **remains current, compatible, and aligned** with new requirements or environments.

Examples:

- The Government recently **changed the VAT rate** from 17.5% to 20%. This change meant that many organizations had to make alterations to their systems.
- Updating a **database system** to work with a **new operating system** or upgraded hardware.
- Updating **security protocols** to align with **new cybersecurity regulations**.
- Adapting a **point-of-sale (POS) system** to support a **new payment gateway** (e.g., eWallet or QR Pay).

Maintenance and Evolution (cont'd.)

Perfective maintenance

- Perfective maintenance involves **making improvements or enhancements** to an existing information system to increase performance, efficiency, or user satisfaction.
- It focuses on **refining the system** based on user feedback or new opportunities, **not because of errors or external changes**, but **to make it work better**.

Examples:

- Improving the **system's response time** or speed of data processing.
- Enhancing the **user interface (UI)** for better usability and appearance.
- Adding **new features** such as search filters or report customization.
- Simplifying **menu navigation** or workflow steps for users.
- Optimizing database queries to handle large datasets more efficiently.
- Adding **data export options** (e.g., to Excel or PDF).

Summary

- **Data** are raw facts
- **Information** is the result of processing data to reveal its meaning
- **Accurate, relevant, and timely information** is the key to good decision making
- Data are usually stored in a **database**
- **DBMS** implements a database and manages its contents
- **Metadata** is data about data
- Databases **evolved from** manual and computerized file systems
- Database management systems were developed to address **file system's inherent weaknesses**
- DBMS present database to end user as single repository (promotes **data sharing**)
- DBMS enforces **data integrity, eliminates redundancy**, and promotes **security**

Summary (cont'd.)

- **Information system** facilitates transformation of data into information
 - **Manages** both data and information
- **SDLC** traces history (life cycle) of an application within the information system
- **DBLC** describes history of database within the information system
- Database design and implementation process moves through **series of well-defined stages**
- **Conceptual design** subject to several variations:
 - Top-down vs. bottom-up